

User Study Report - Usability & Trust Evaluation

The objective of this user study was to evaluate the system usability of the VerifAI deepfake detection interface using the System Usability Scale (SUS), as well as to assess user trust and confidence in AI-generated deepfake predictions, informed by trust-related metrics described in Sippy et al. (2024). This study was conducted as part of the human-centred evaluation phase of the VerifAI framework, complementing prior technical and cross-dataset performance analyses.

The study involved a total of 30 participants who interacted with the VerifAI system over two consecutive days, with 15 participants completing the study on 31 January 2026 and the remaining 15 on 1 February 2026. Participants were grouped into two age ranges, 18-30 and 31-60, and all sessions were conducted using my personal laptop to ensure consistency in interaction conditions. Ethical considerations were addressed through the provision of a participant information sheet, informed consent form, task instructions, and a post-study debrief. All collected data was anonymised, and no personally identifiable information was stored.

Participants were asked to complete a predefined interaction task involving the use of the VerifAI system to inspect deepfake detection outputs and model confidence scores. During the task, interaction metrics such as task completion time, task success, and error counts were recorded. Following task completion, participants completed a post-task questionnaire consisting of the ten SUS items and additional questions related to trust in the system, perceived reliability, and prior awareness of deepfake technology.

The collected responses were processed using an automated analysis script that computed individual SUS scores on a 0-100 scale, along with an aggregated trust index derived from user-reported confidence measures. The resulting SUS score distribution indicated a wide but realistic range of usability perceptions, with most participants scoring the system within the “good usability” range (70-80). Lower scores were associated with longer task completion times and higher error counts, while higher scores corresponded to faster task completion and fewer interaction issues. This variation reflects expected differences in user familiarity and technical confidence.

Trust-related results showed moderate to high confidence in the system’s predictions for the majority of participants, although trust levels varied independently from usability scores in some cases. This finding aligns with existing research suggesting that system usability and trust, while related, represent distinct dimensions of user experience. Overall, participants demonstrated an ability to engage with the system effectively, and the results suggest that VerifAI provides a usable and reasonably trustworthy interface for deepfake detection tasks.

In conclusion, this user study provides evidence that the VerifAI framework achieves satisfactory usability and user trust in a realistic interaction setting. These findings support the feasibility of the system for practical use and highlight the importance of combining technical performance evaluation with human-centred assessment. The results obtained in this study form a valuable foundation for future, larger-scale user evaluations and further refinement of the system's interface and explainability features.